

ICSA 2026 SHORT COURSE

Lecture 1: Introduction

Statistical Foundations of Transfer Learning

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Overview

§1.1: Why, when, and where

§1.2: Concepts

§1.3: Roadmap of the short course

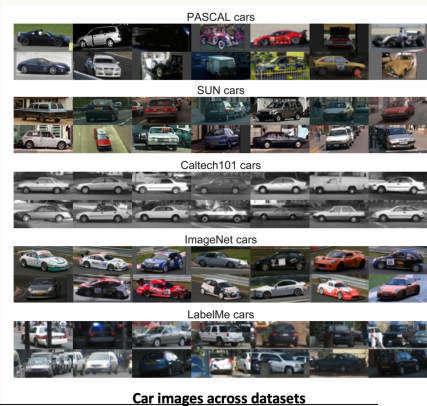
References

§1.1: Why, when, and where

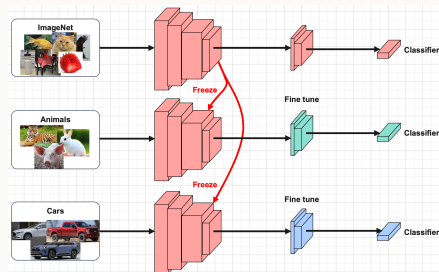
Transfer learning is everywhere

“You trained a model. Now it has to work somewhere else. What do you do?”

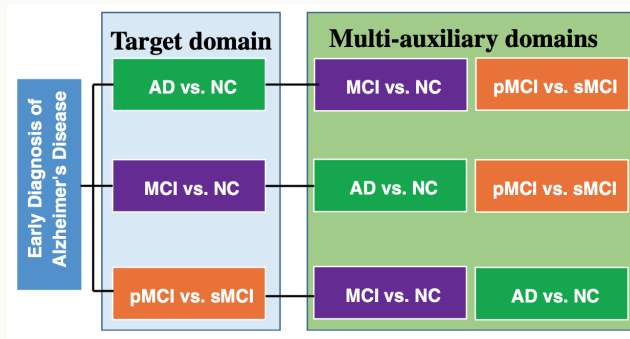
Computer vision



Large language models



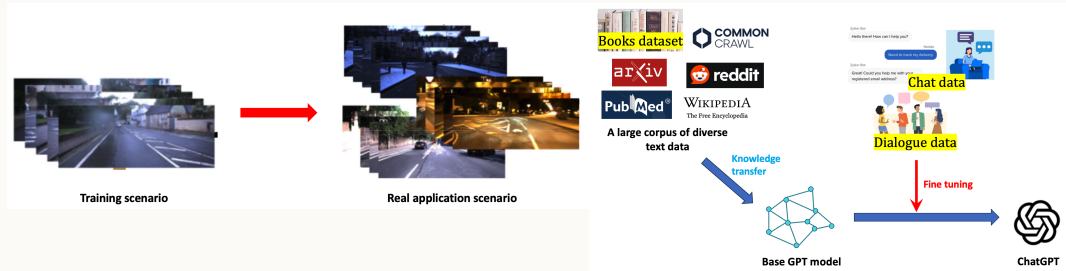
Transfer learning in healthcare



AD: Alzheimer's; MCI: mild cognitive impairment (pMCI / sMCI); NC: normal controls.

Cheng et al. (2017). Multi-domain transfer learning for early diagnosis of Alzheimer's disease. *Neuroinformatics*, 15.

Transfer learning in autonomous driving & chat



Akhauri, Zheng, Lin (2020). Enhanced transfer learning for autonomous driving. *IROS*.

Why, when, and where

- **Why:** improve performance on a target task with limited (or zero) labeled data.
- **When:** when source and target are *related enough* that knowledge can transfer.
- **Where:** computer vision, NLP, healthcare, finance, autonomous driving, ...

See Pan and Yang (2009); Weiss et al. (2016); Zhuang et al. (2020) for comprehensive surveys.

Our running example: Alzheimer's from brain MRI

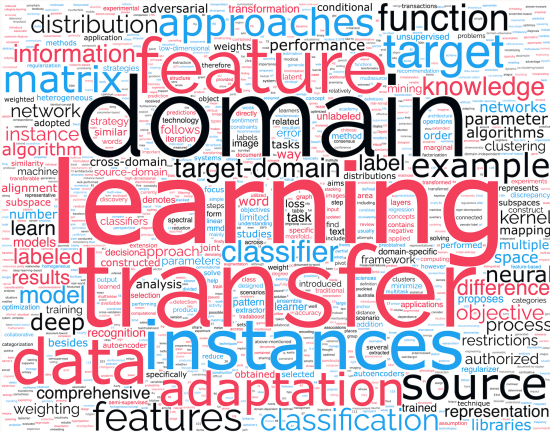
The scenario

Train an Alzheimer's classifier on **500 research-study MRIs** (the *source* domain), then deploy it at a **community hospital** that has only **20 labeled scans** (the *target* domain).

- **Why transfer?** Training on the hospital's 20 scans alone gives $\approx 30\%$ error: far too few labels to learn a reliable classifier from scratch, so we must borrow strength from the 500 source scans.
- **When is it justified?** The underlying *disease biology is the same* at both sites, so what the model learned from the source is genuinely relevant to the target $\Rightarrow$ transfer is warranted.
- **Where does this live?** Neuroimaging — but the same “data-rich source, data-poor target” pattern recurs in vision, NLP, finance, ...

We revisit this example as each new concept (covariate shift, posterior drift, ...) appears.

§1.2: Concepts



Papers say domain adaptation, distributional robustness, fine-tuning, dataset shift — all instances of the framework we formalize next.

Word cloud generated from Zhuang et al. (2020), *Proceedings of the IEEE*, 109(1).

Domain, task, transfer

- **Domain** $(\mathcal{S}^{(k)}, \mathbb{P}^{(k)})$: a sample space $\mathcal{S}^{(k)} = \mathcal{X} \times \mathcal{Y}$ with *joint* law $\mathbb{P}^{(k)}$. Index $k = 0$ for *target*, $k = 1, \dots$ for *sources*. (which factor of $\mathbb{P}^{(k)} = \mathbb{P}_X^{(k)} \mathbb{P}_{Y|X}^{(k)}$ moves — detailed shortly) (e.g., the joint law of MRI pixels and disease labels at the community-hospital target)
- **Task** $t^{(k)}$: the learning goal on the domain — a predictor $\mathcal{X} \rightarrow \mathcal{Y}$ and loss ℓ (classification, regression, ...). (e.g., predict Alzheimer's from MRI)
- **Transfer learning**: improve performance on the target task $t^{(0)}$ using knowledge of source task(s) $t^{(1)}, \dots, t^{(K)}$. (e.g., use the 500 research-study source MRIs to build a better classifier for the community hospital's 20 target scans)

Most definitions follow Redko et al. (2019, 2020).

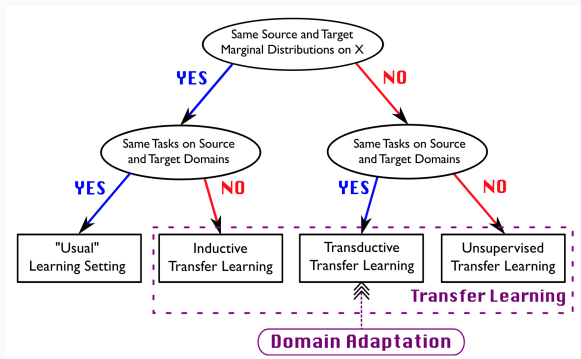
Categories by goal: TL vs. MTL

- **Transfer learning (TL):** use source(s) to improve performance on a single *target*.
- **Multi-task learning (MTL):** treat each domain as a *task*; learn well on *all* tasks simultaneously by sharing structure across them.

Same machinery (joint estimation, shared regularization), different objective.

Categories by task type

This course focuses on **homogeneous** transfer (shared $\mathcal{X}, \mathcal{Y}$; same task, shifted distribution) — the left branches below. Heterogeneous transfer lets $\mathcal{X}$ or $\mathcal{Y}$ differ (e.g. text source $\rightarrow$ image target), needing a feature-space map; we restrict to **homogeneous** transfer (shared $\mathcal{X}, \mathcal{Y}$), where theory is tractable and fits most health/tabular settings.



Categories by shift type

Joint distribution $\mathbb{P}^{(k)}$ on $\mathcal{X} \times \mathcal{Y}$, target $k = 0$, source $k = 1$:

$$\text{Any joint shift decomposes: } \mathbb{P}^{(k)} = \underbrace{\mathbb{P}_X^{(k)}}_{\text{marginal}} \cdot \underbrace{\mathbb{P}_{Y|X}^{(k)}}_{\text{conditional}}.$$

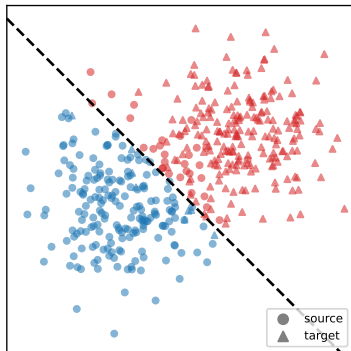
- **Covariate shift:** $\mathbb{P}_X^{(0)} \neq \mathbb{P}_X^{(1)}$, $\mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)}$ (a.s.)
- **Posterior drift:** $\mathbb{P}_X^{(0)} = \mathbb{P}_X^{(1)}$, $\mathbb{P}_{Y|X}^{(0)} \neq \mathbb{P}_{Y|X}^{(1)}$
- **Concept drift:** $\mathbb{P}_X^{(0)} \neq \mathbb{P}_X^{(1)}$, $\mathbb{P}_{Y|X}^{(0)} \neq \mathbb{P}_{Y|X}^{(1)}$

MRI example: covariate shift = a new scanner at the target site; posterior drift = diagnosis criteria differ across sites; concept drift = both at once (the hardest case).

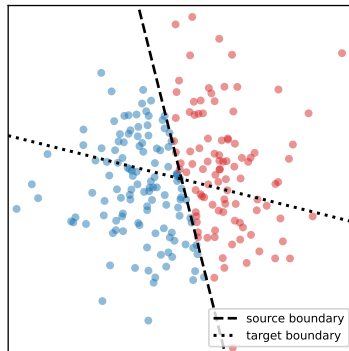
Key question: fix x — does $\mathbb{P}_{Y|X}$ change? Yes $\Rightarrow$ posterior/concept drift. No $\Rightarrow$ covariate shift.

Categories by shift type (cont.)

Covariate shift: P_X moves, boundary shared

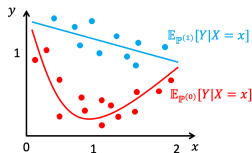
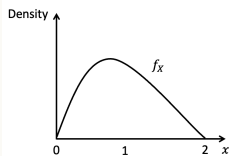


Posterior drift: P_X shared, boundary rotates

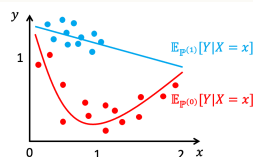
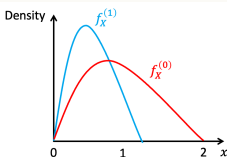


Posterior drift and concept drift, visualized

Posterior drift



Concept drift



Posterior drift: the boundary moves while P_X stays fixed. Concept drift: input cloud and boundary both shift — the hardest case.

Check-in: which kind of shift?

A model is trained on MRI scans collected in a research study and deployed in a community hospital using different scanner manufacturers. Disease prevalence in both populations is the same, but scanner-induced features look statistically different.

Which kind of shift is this?

- (a) Covariate shift: $\mathbb{P}_X^{(0)} \neq \mathbb{P}_X^{(1)}, \mathbb{P}_{Y|X}^{(0)} = \mathbb{P}_{Y|X}^{(1)}$
- (b) Posterior drift: $\mathbb{P}_X^{(0)} = \mathbb{P}_X^{(1)}, \mathbb{P}_{Y|X}^{(0)} \neq \mathbb{P}_{Y|X}^{(1)}$
- (c) Concept drift: both differ
- (d) No shift

Answer

(a) **Covariate shift.** The marginal $\mathbb{P}_X$ changes (scanner-induced feature distributions differ), but the disease-given-features mechanism $\mathbb{P}_{Y|X}$ is unchanged because biology is the same. **Consequence:** importance-reweighting (L3) can fix covariate shift; if it were posterior drift, reweighting alone would not.

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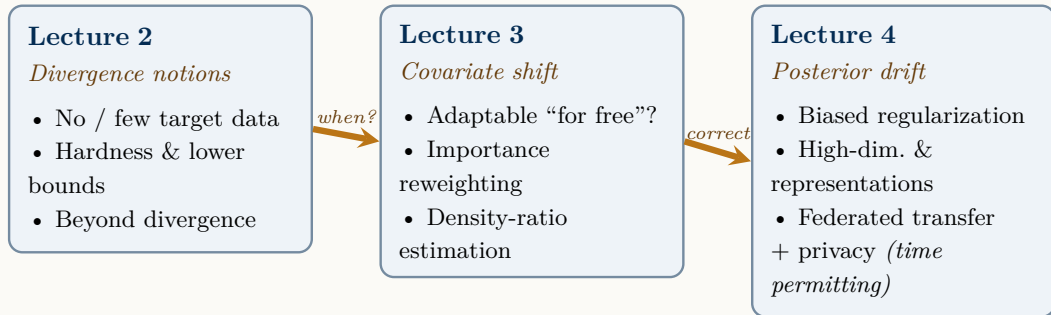
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§1.3: Roadmap of the short course

Roadmap: three threads

Recap of today: vocabulary (domain / task / transfer), the three shift types (covariate / posterior / concept), and our running MRI example. Next: making “related enough” precise.



Roadmap: three threads (cont.)

- **Divergence-based bounds (L2):** when is transfer possible at all?
- **Covariate shift & reweighting (L3):** correcting a moving P_X .
- **Posterior drift, biased regularization & representations (L4):** safe transfer of the model.

Order follows dependency: know when transfer helps (L2) before designing methods that correct for shift (L3–L4).

References

References

- Pan, S. J. and Yang, Q. (2009). A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*, 22(10):1345–1359.
- Redko, I., Morvant, E., Habrard, A., Sebban, M., and Bennani, Y. (2019). *Advances in domain adaptation theory*. Elsevier.
- Redko, I., Morvant, E., Habrard, A., Sebban, M., and Bennani, Y. (2020). A survey on domain adaptation theory: learning bounds and theoretical guarantees. *arXiv preprint arXiv:2004.11829*.
- Weiss, K., Khoshgoftaar, T. M., and Wang, D. (2016). A survey of transfer learning. *Journal of Big data*, 3(1):1–40.
- Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., and He, Q. (2020). A comprehensive survey on transfer learning. *Proceedings of the IEEE*, 109(1):43–76.